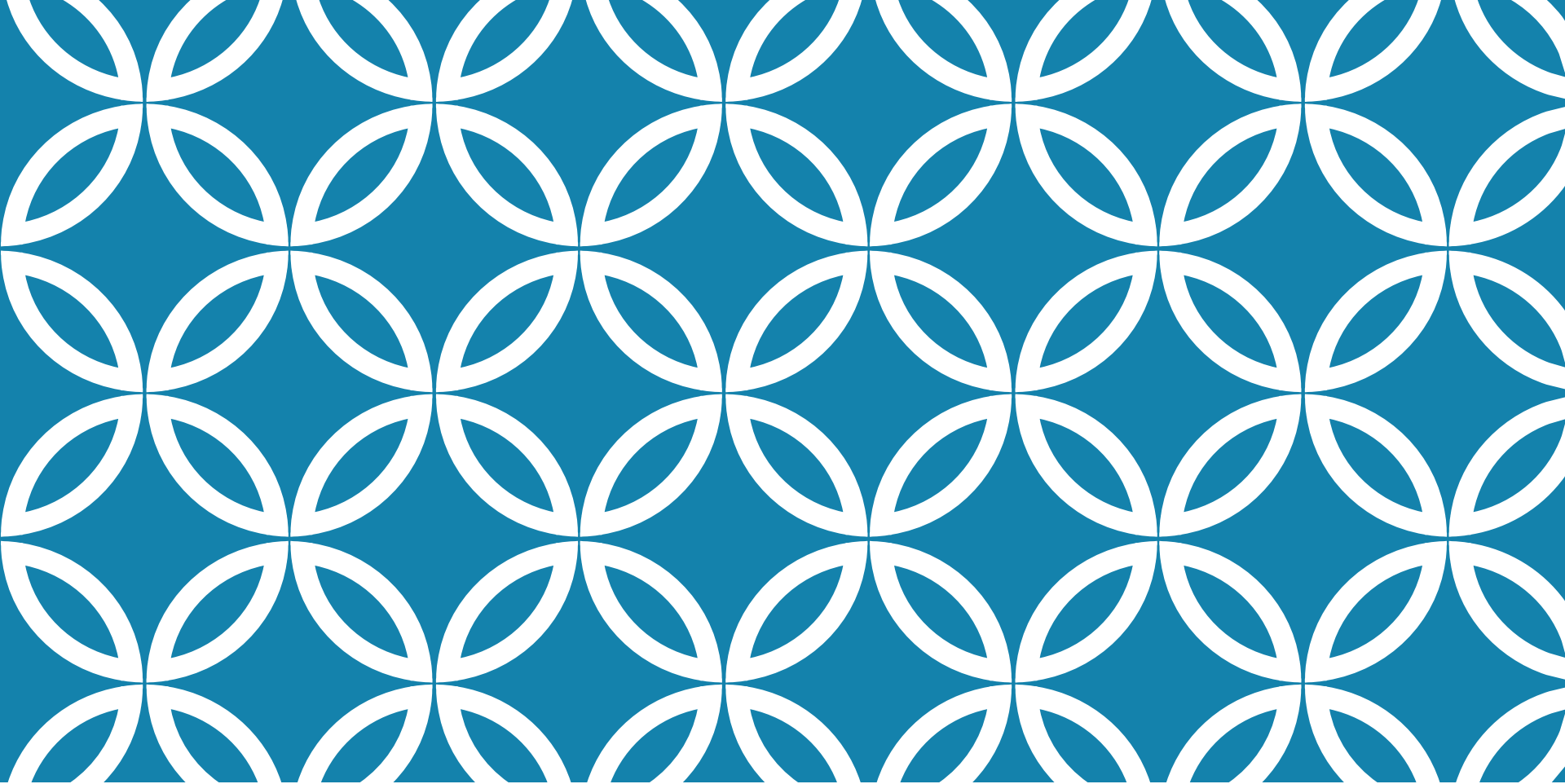




IT SOCIETY

INTRODUCTION

DATABASES

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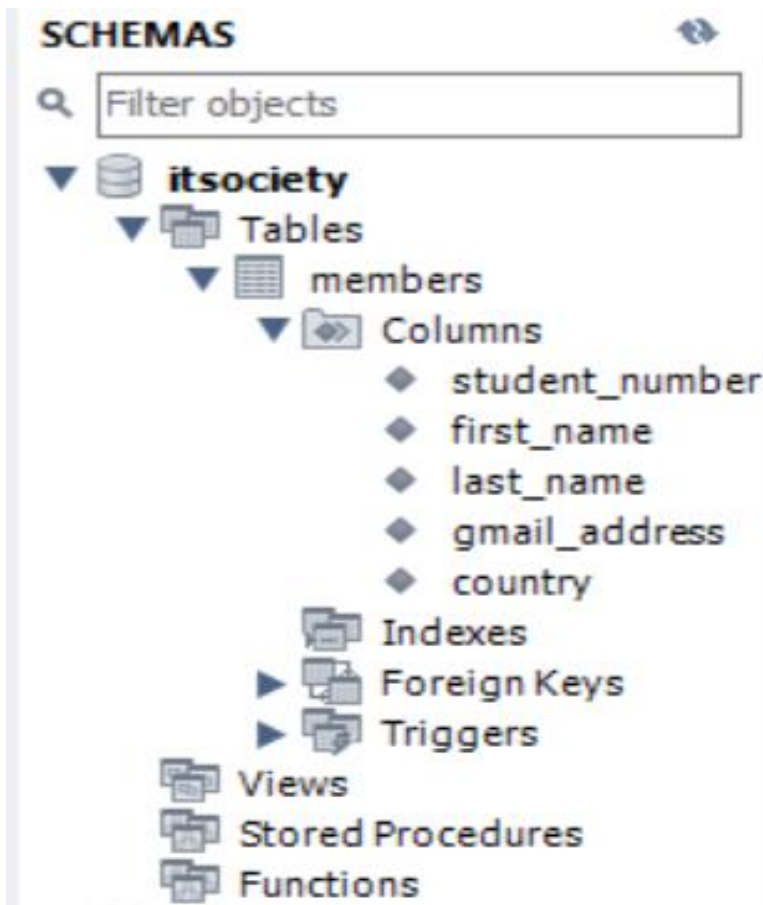
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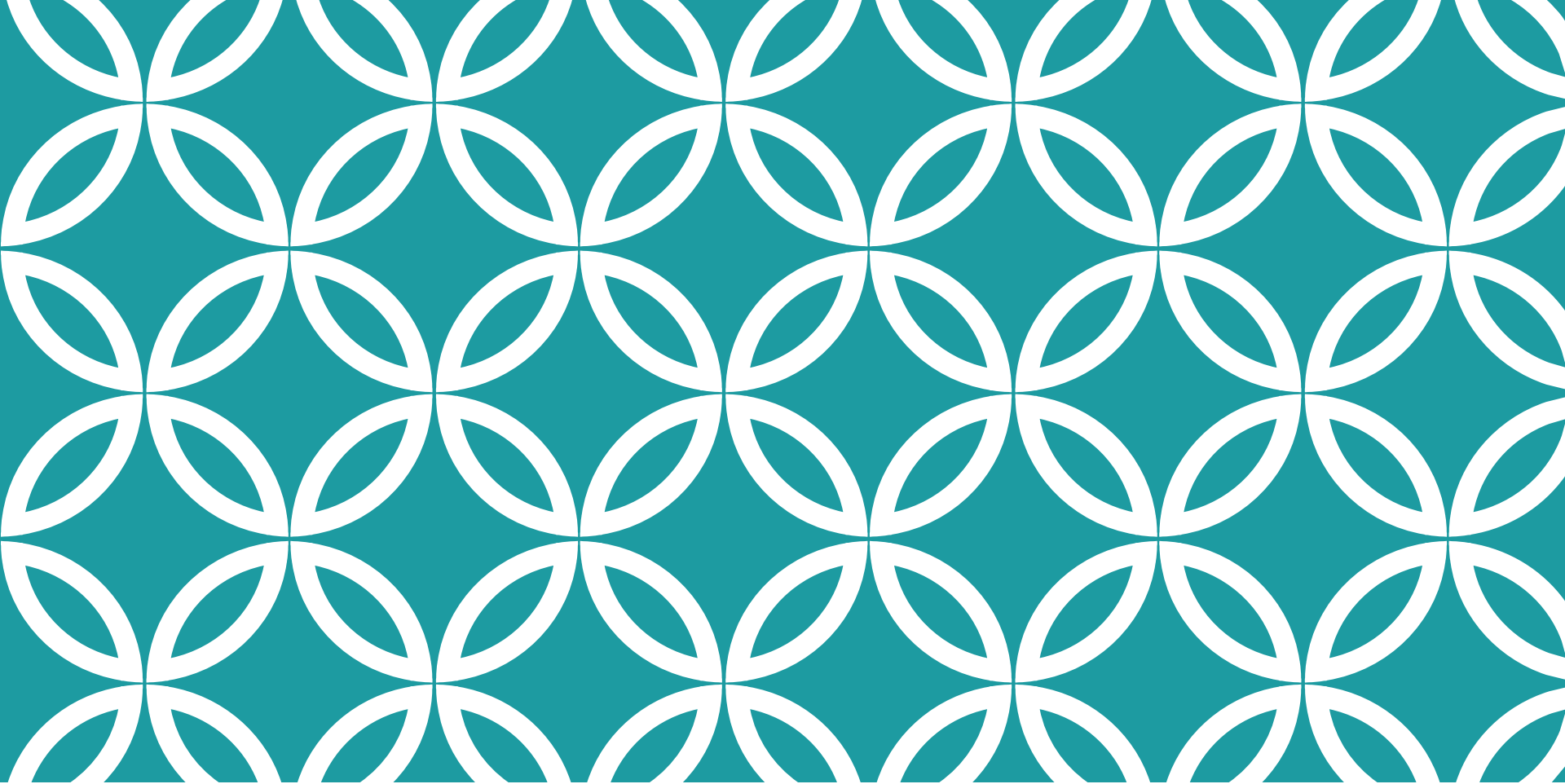
RELATIONAL DATABASES



INTRODUCTION



- A **database schema** defines how data is organized within a relational database.
- In a relational database, **columns** are called **fields** and **rows** are called **records**.



DATA MANIPULATION LANGUAGE

INSERT, UPDATE & DELETE

INTRODUCTION

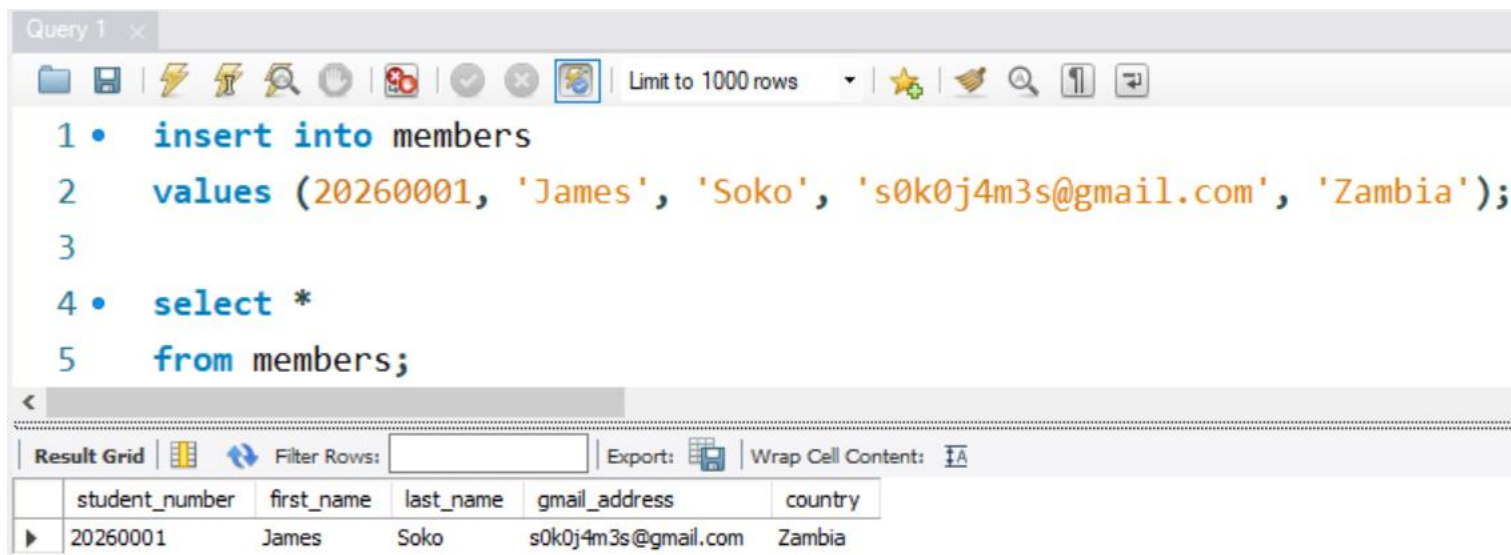
- **DML** commands are used to manipulate data within a database.
- They allow you to **insert, update, delete, and retrieve** data from tables.
- The include;
 1. SELECT
 2. INSERT
 3. UPDATE
 4. DELETE

3. INSERT INTO

VARIATION	PURPOSE
Single Row	Insert one record
Multiple Rows	Insert several records in one statement
All Columns	Omit column list; must provide values for every column in table order
SET Syntax	Alternative syntax, inserts one row
From SELECT	Copy rows from another table

INSERT INTO | ALL COLUMNS & SINGLE ROW

- With this method, you don't need to list any field but if no record of the field is missing its an error, so all records for each field must be present.



The screenshot shows a database query editor window titled "Query 1". The query text is as follows:

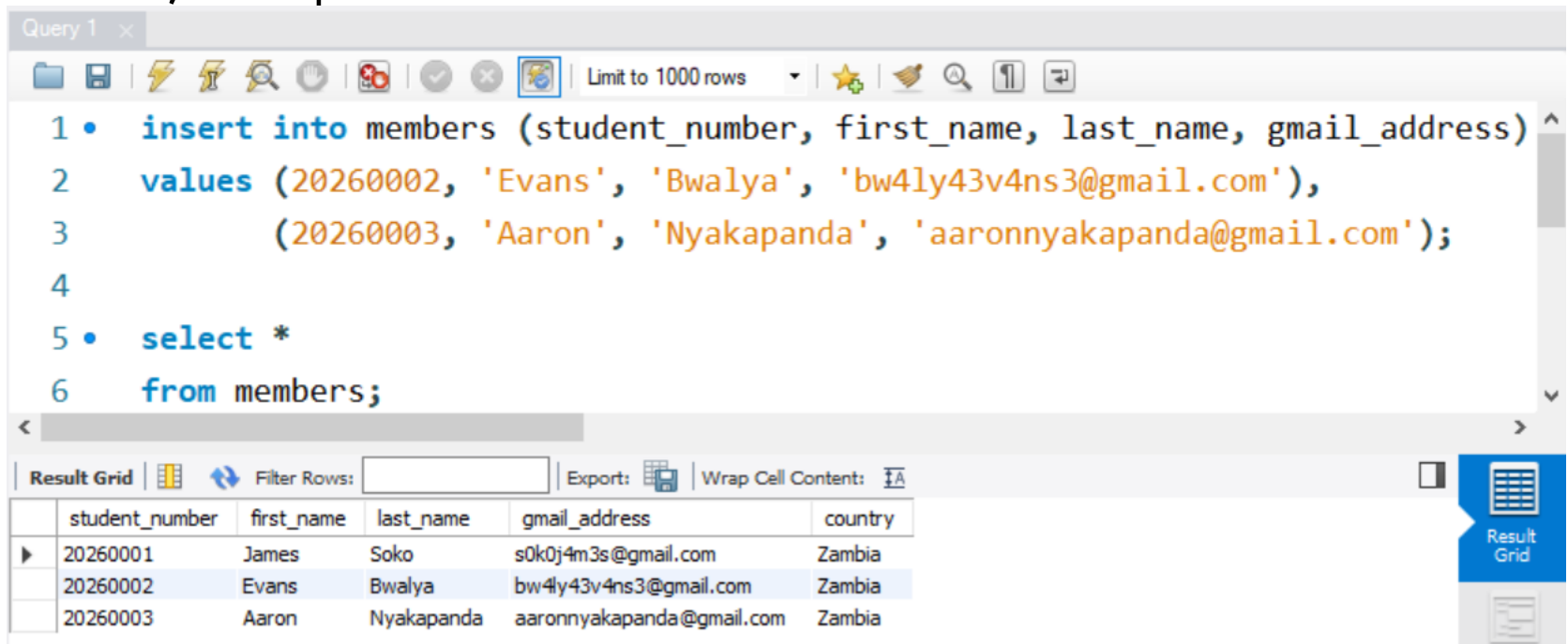
```
1 • insert into members
2 values (20260001, 'James', 'Soko', 's0k0j4m3s@gmail.com', 'Zambia');
3
4 • select *
5 from members;
```

Below the query editor, the "Result Grid" is displayed, showing the result of the query. The grid has the following columns: student_number, first_name, last_name, gmail_address, and country. The first row of data is:

student_number	first_name	last_name	gmail_address	country
20260001	James	Soko	s0k0j4m3s@gmail.com	Zambia

INSERT INTO | SPECIFIED COLUMNS & MULTIPLE ROWS

- For this one, will specify the target fields are attempt to insert multiple records, recall the country field as a default constraint of Zambia, we expect that to be auto-fill.



The screenshot shows a database query editor interface. The query editor displays the following SQL code:

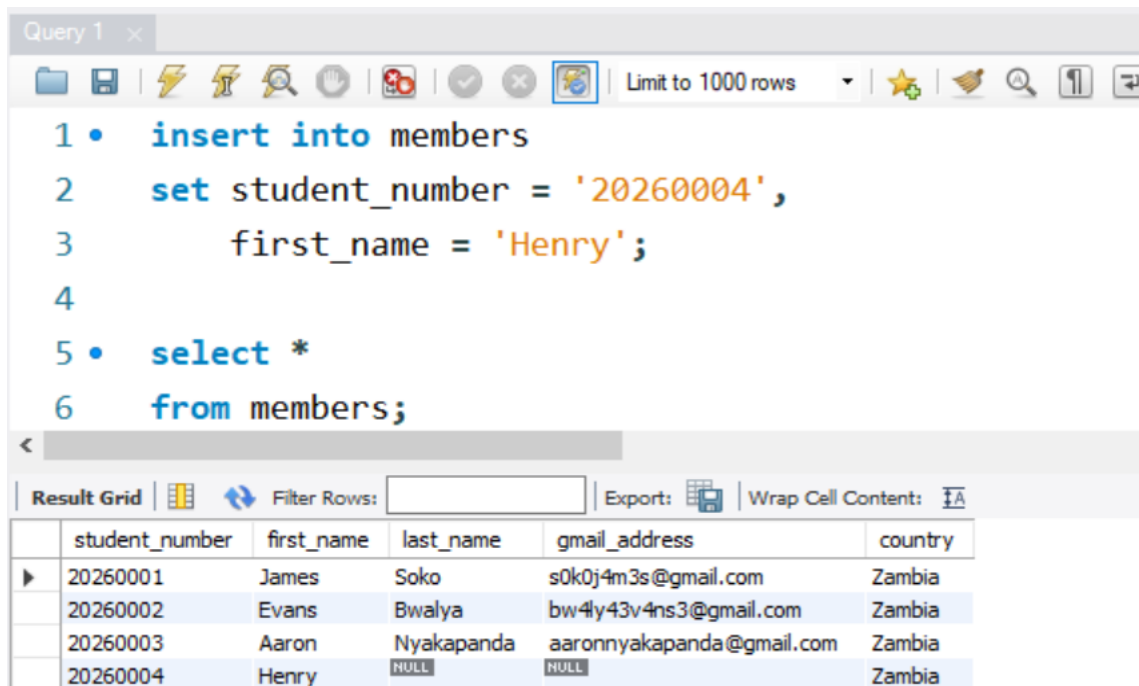
```
1 • insert into members (student_number, first_name, last_name, gmail_address)
2 values (20260002, 'Evans', 'Bwalya', 'bw4ly43v4ns3@gmail.com'),
3        (20260003, 'Aaron', 'Nyakapanda', 'aaronnyakapanda@gmail.com');
4
5 • select *
6 from members;
```

Below the query editor, the 'Result Grid' is visible, showing the results of the query. The grid has columns: student_number, first_name, last_name, gmail_address, and country. The results are as follows:

student_number	first_name	last_name	gmail_address	country
20260001	James	Soko	s0k0j4m3s@gmail.com	Zambia
20260002	Evans	Bwalya	bw4ly43v4ns3@gmail.com	Zambia
20260003	Aaron	Nyakapanda	aaronnyakapanda@gmail.com	Zambia

INSERT INTO | SET SYNTAX

- For this one, we don't need the VALUES command.



The screenshot shows a database query editor window titled "Query 1". The query is as follows:

```
1 • insert into members
2   set student_number = '20260004',
3     first_name = 'Henry';
4
5 • select *
6   from members;
```

Below the query, the "Result Grid" shows the data returned by the SELECT statement. The table has columns: student_number, first_name, last_name, gmail_address, and country. The data is as follows:

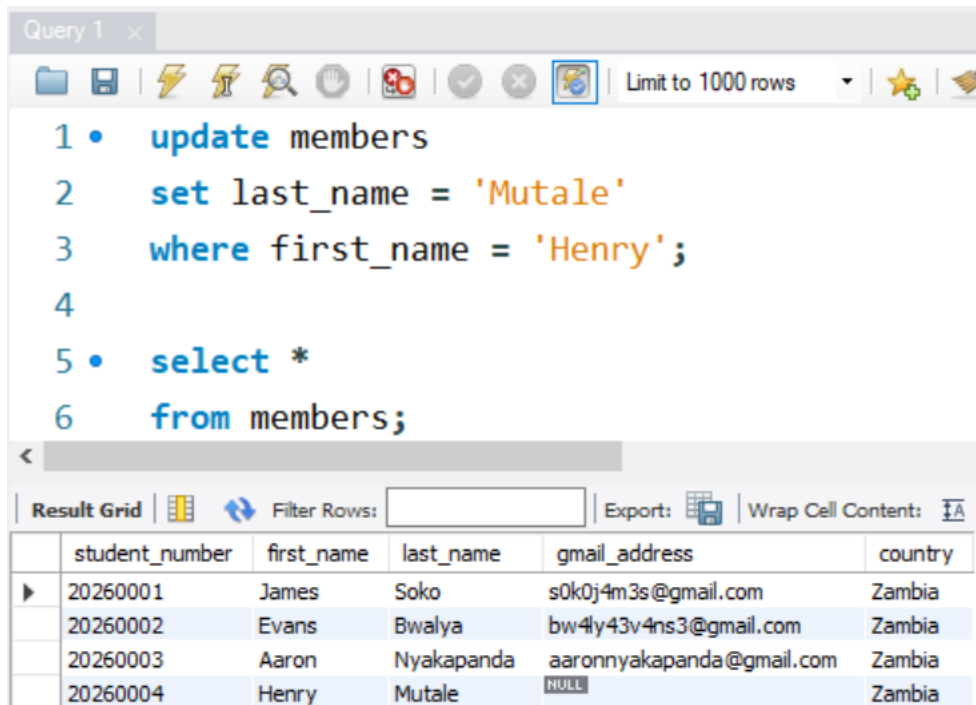
	student_number	first_name	last_name	gmail_address	country
▶	20260001	James	Soko	s0k0j4m3s@gmail.com	Zambia
	20260002	Evans	Bwalya	bw4ly43v4ns3@gmail.com	Zambia
	20260003	Aaron	Nyakapanda	aaronnyakapanda@gmail.com	Zambia
	20260004	Henry	NULL	NULL	Zambia

4. UPDATE

VARIATION	PURPOSE
Single Field	Used to update a single record of the specified field.
Multiple Fields	Used to update multiple records of the specified fields.

UPDATE | SINGLE FIELD

- For this one, we can just target the last_name field;



The screenshot shows a database query editor interface. At the top, there's a tab labeled 'Query 1'. Below the tab is a toolbar with various icons for file operations, execution, and viewing. The main area contains a SQL query:

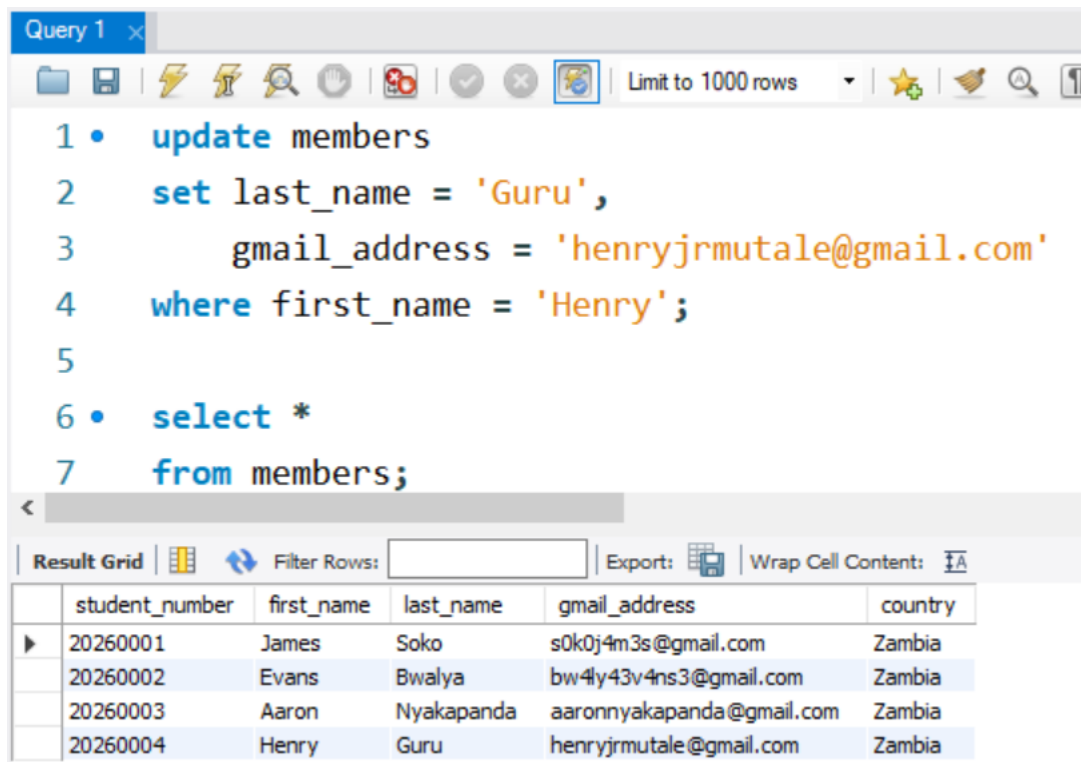
```
1 • update members
2   set last_name = 'Mutale'
3   where first_name = 'Henry';
4
5 • select *
6   from members;
```

Below the query, there's a 'Result Grid' section. It includes a 'Filter Rows' input field, an 'Export' button, and a 'Wrap Cell Content' checkbox. The result grid displays the following data:

	student_number	first_name	last_name	gmail_address	country
▶	20260001	James	Soko	s0k0j4m3s@gmail.com	Zambia
	20260002	Evans	Bwalya	bw4ly43v4ns3@gmail.com	Zambia
	20260003	Aaron	Nyakapanda	aaronnyakapanda@gmail.com	Zambia
	20260004	Henry	Mutale	NULL	Zambia

UPDATE | MULTIPLE FIELDS

- For this one, we can target more than one field;



The screenshot shows a database query editor interface. The query window displays the following SQL code:

```
1 • update members
2   set last_name = 'Guru',
3       gmail_address = 'henryjrmutale@gmail.com'
4   where first_name = 'Henry';
5
6 • select *
7   from members;
```

Below the query window, the 'Result Grid' is visible, showing the results of the SELECT statement. The grid has 6 columns: student_number, first_name, last_name, gmail_address, and country. The first row is highlighted with a mouse cursor.

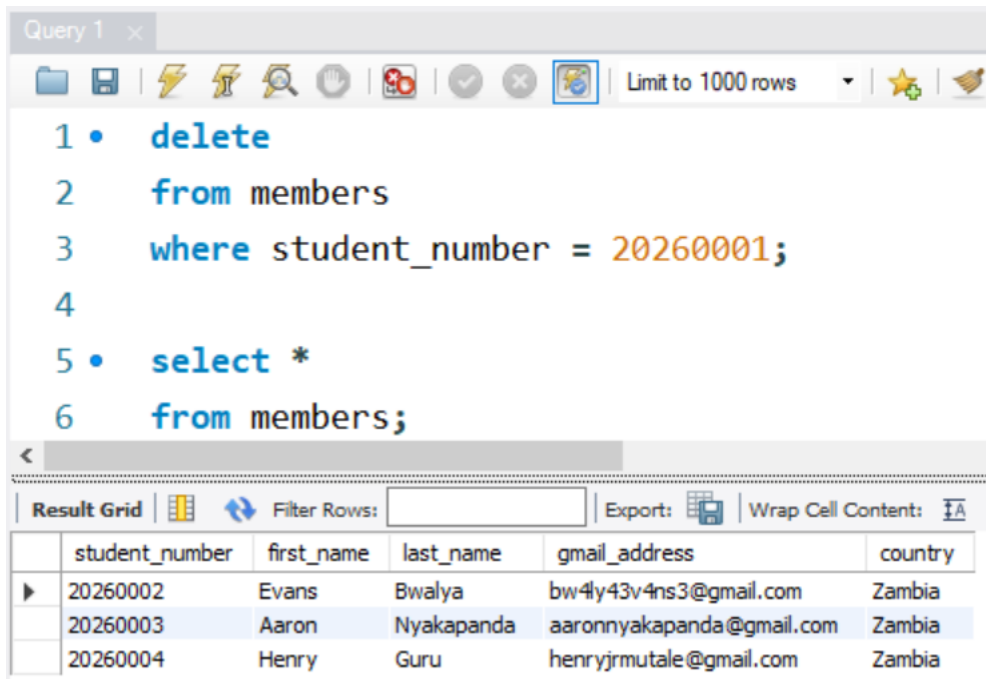
	student_number	first_name	last_name	gmail_address	country
▶	20260001	James	Soko	s0k0j4m3s@gmail.com	Zambia
	20260002	Evans	Bwalya	bw4ly43v4ns3@gmail.com	Zambia
	20260003	Aaron	Nyakapanda	aaronnyakapanda@gmail.com	Zambia
	20260004	Henry	Guru	henryjrmutale@gmail.com	Zambia

5. DELETE

VARIATION	PURPOSE
Single Row	Removes a single row
Multiple Rows	Removes the specified rows
All Rows	Remove every row (table structure remains)

DELETE | SINGLE ROW

- You've just realized that James is the IT Society Technical Team Leader, hence his data must not be found in the table of members of the society.



The screenshot shows a database query editor interface. The top toolbar includes icons for file operations, query execution, and a 'Limit to 1000 rows' dropdown. The query editor contains the following SQL code:

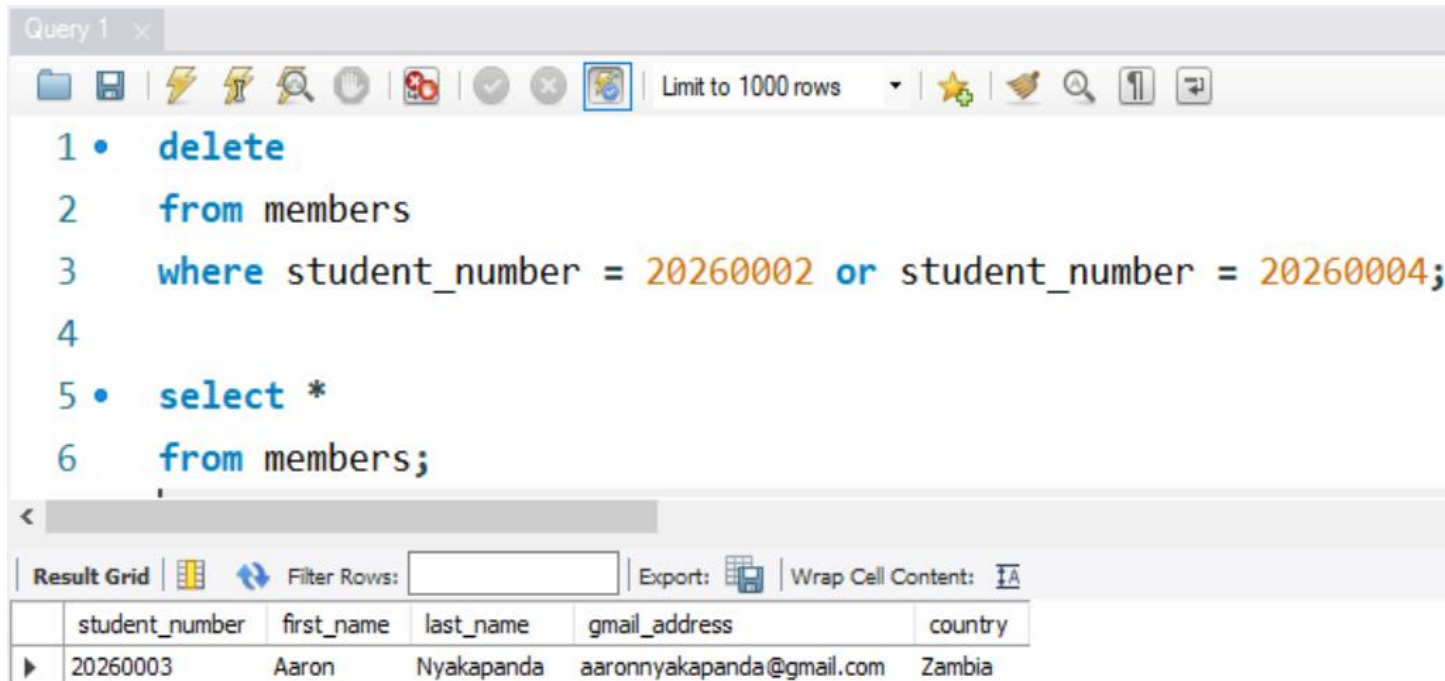
```
1 • delete
2   from members
3   where student_number = 20260001;
4
5 • select *
6   from members;
```

Below the query editor is a 'Result Grid' tab. It features a 'Filter Rows' input field, an 'Export' button, and a 'Wrap Cell Content' checkbox. The grid displays the following data:

	student_number	first_name	last_name	gmail_address	country
▶	20260002	Evans	Bwalya	bw4ly43v4ns3@gmail.com	Zambia
	20260003	Aaron	Nyakapanda	aaronnyakapanda@gmail.com	Zambia
	20260004	Henry	Guru	henryjrmutale@gmail.com	Zambia

DELETE | MULTIPLE ROWS

- Oh, Evans and Henry are part of the Executive, let's remove their records in the members table.



The screenshot shows a database query editor interface. The top toolbar includes icons for file operations, execution, and filtering, with a 'Limit to 1000 rows' dropdown. The query editor contains the following SQL code:

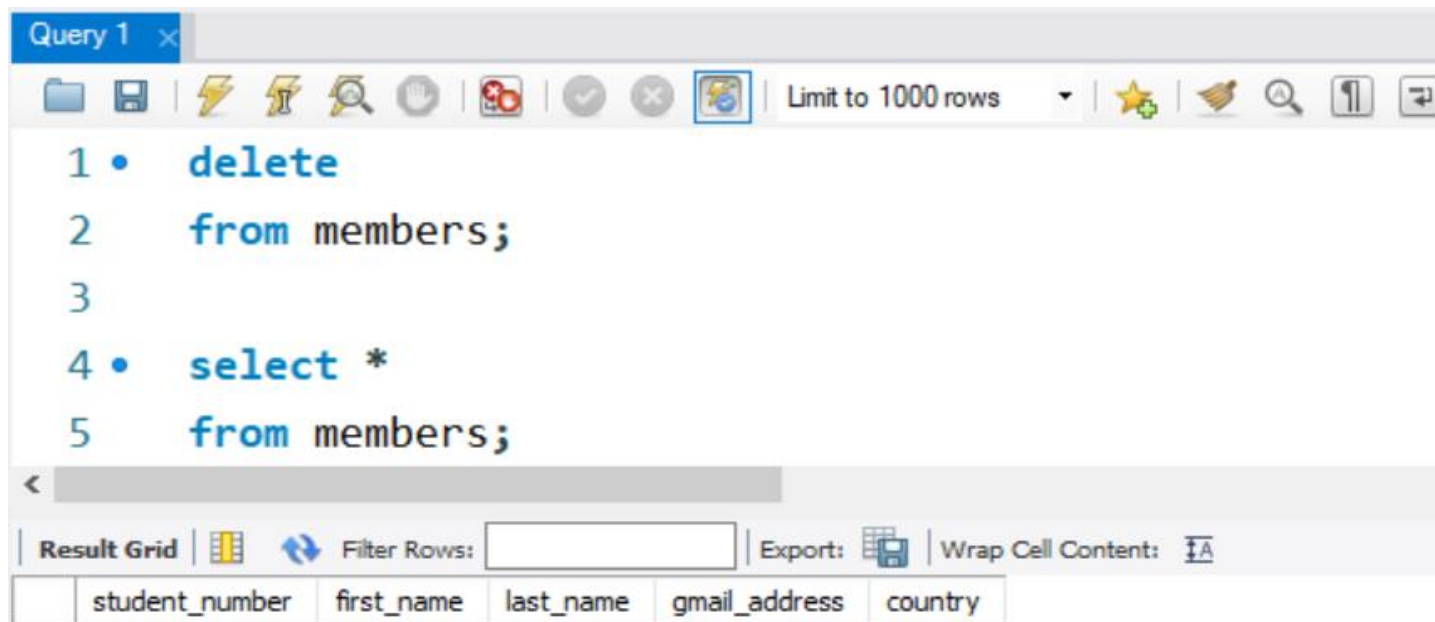
```
1 • delete
2   from members
3   where student_number = 20260002 or student_number = 20260004;
4
5 • select *
6   from members;
```

Below the query editor is a 'Result Grid' section. It includes a 'Filter Rows' input field, an 'Export' button, and a 'Wrap Cell Content' checkbox. The grid displays one row of data from the 'members' table:

	student_number	first_name	last_name	gmail_address	country
▶	20260003	Aaron	Nyakapanda	aaronnyakapanda@gmail.com	Zambia

DELETE | ALL ROWS

- Only run this command if you want to get rid of all records.



The screenshot shows a database query editor window titled "Query 1". The toolbar includes icons for file operations, execution, and a "Limit to 1000 rows" dropdown. The query text is as follows:

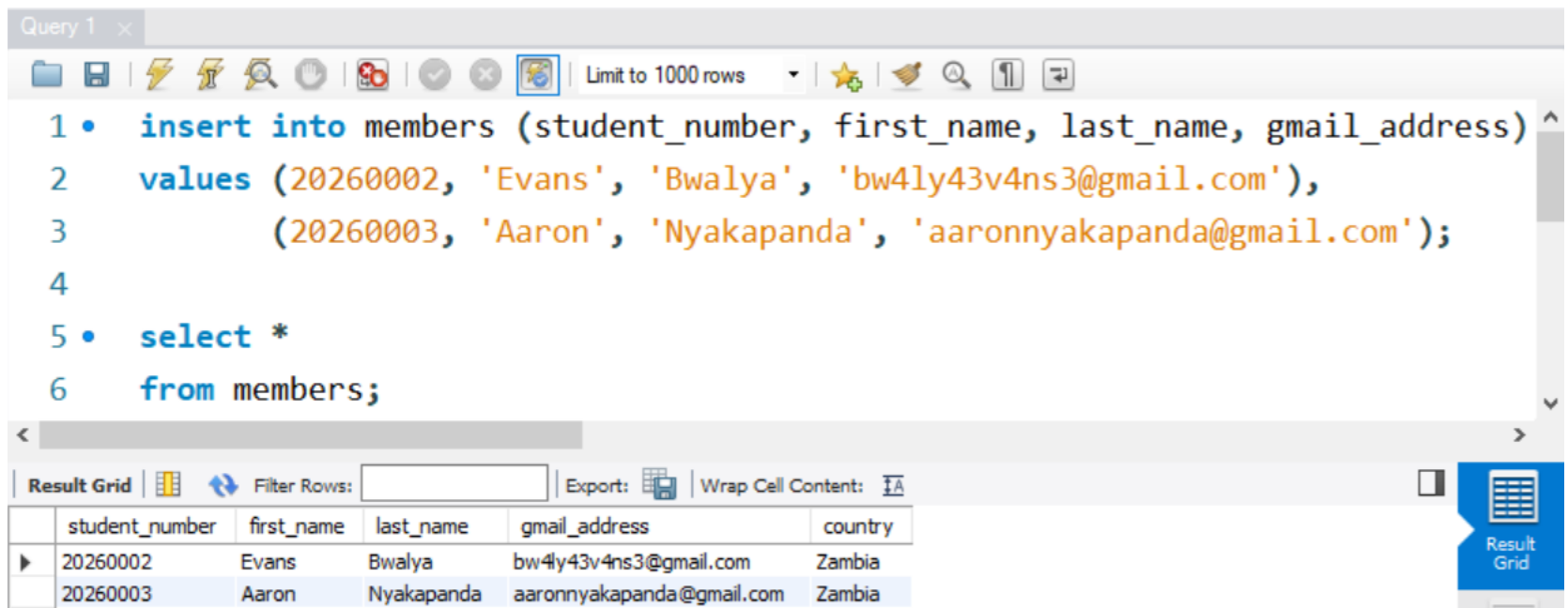
```
1 • delete
2   from members;
3
4 • select *
5   from members;
```

Below the query editor, there is a "Result Grid" section with a "Filter Rows" input field and an "Export" button. The result grid displays the following columns:

student_number	first_name	last_name	gmail_address	country
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6. TRUNCATE

- The best way to delete all records is by using the TRUNCATE command, for demonstration, let's insert the records as shown below;



The screenshot shows a database query editor window titled 'Query 1'. The SQL code is as follows:

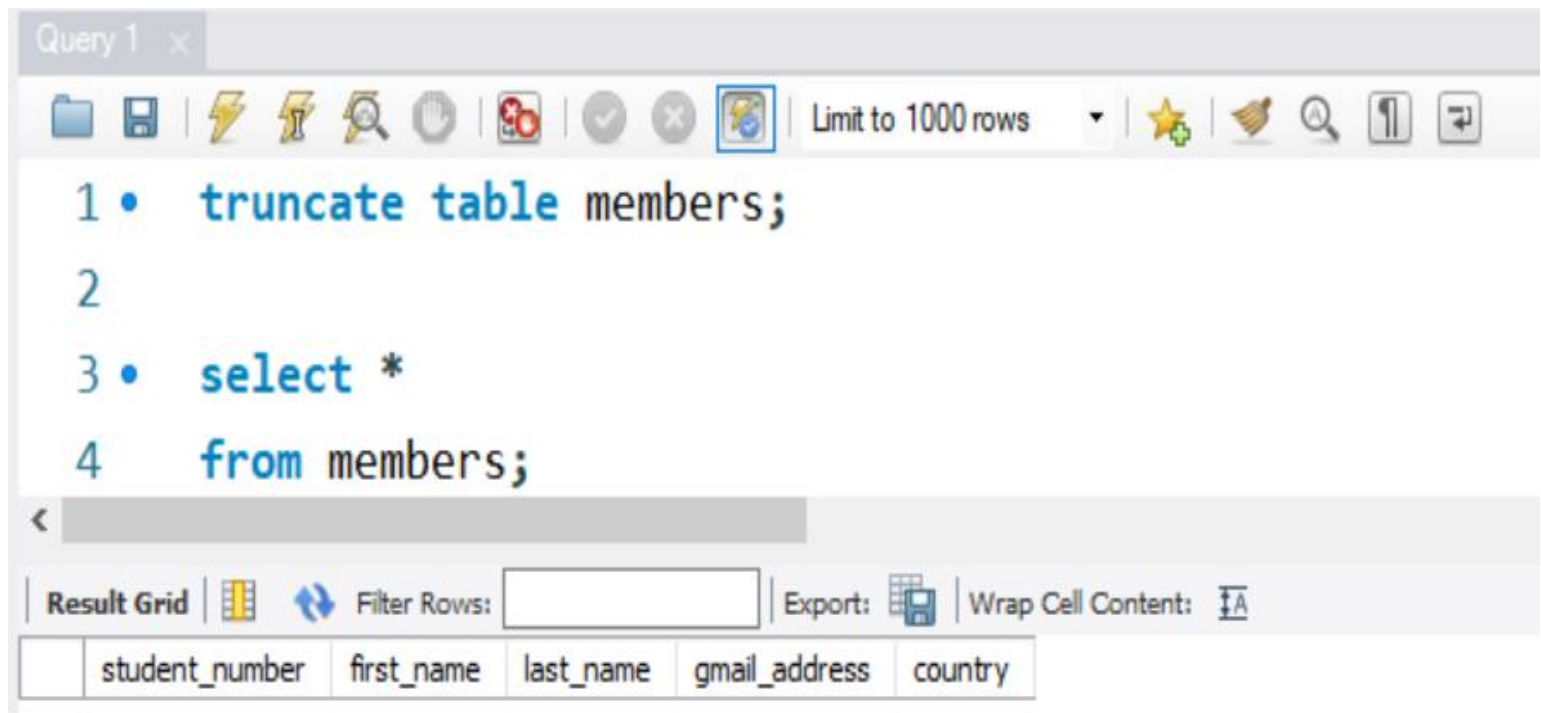
```
1 • insert into members (student_number, first_name, last_name, gmail_address)
2 values (20260002, 'Evans', 'Bwalya', 'bw4ly43v4ns3@gmail.com'),
3        (20260003, 'Aaron', 'Nyakapanda', 'aaronnyakapanda@gmail.com');
4
5 • select *
6 from members;
```

Below the query editor, the 'Result Grid' is displayed, showing the results of the SELECT query. The grid has columns for student_number, first_name, last_name, gmail_address, and country. The results are as follows:

student_number	first_name	last_name	gmail_address	country
20260002	Evans	Bwalya	bw4ly43v4ns3@gmail.com	Zambia
20260003	Aaron	Nyakapanda	aaronnyakapanda@gmail.com	Zambia

TRUNCATE (CONTINUES...)

- Get rid of all records.



CONCLUSION

- At this point, we now understand what is a database **schema**, **field** and **record**.
- In this second handout, our focus was to understand some uses of the **INSERT INTO**, **UPDATE**, **DELETE** and **TRUNCATE** commands, as we progress will continue using these commands by introducing new things.



UP NEXT... TRANSACTION CONTROL LANGUAGE